JEE-MAIN **Date: 12-03-2025**
FST-05 **Max. Marks: 300**

KEY SHEET

PHYSICS

1)	**2**	2)	**3**	3)	**3**	4)	**1**	5)	**1**
6)	**1**	7)	**4**	8)	**3**	9)	**2**	10)	**2**
11)	**3**	12)	**4**	13)	**2**	14)	**4**	15)	**2**
16)	**2**	17)	**3**	18)	**1**	19)	**2**	20)	**3**
21)	**4**	22)	**3**	23)	**2**	24)	**8**	25)	**7**
26)	**4**	27)	**7**	28)	**6**	29)	**9**	30)	**6**

CHEMISTRY

31)	**1**	32)	**4**	33)	**1**	34)	**4**	35)	**2**
36)	**1**	37)	**2**	38)	**4**	39)	**3**	40)	**2**
41)	**1**	42)	**1**	43)	**2**	44)	**4**	45)	**2**
46)	**1**	47)	**4**	48)	**1**	49)	**1**	50)	**2**
51)	**325**	52)	**2**	53)	**3**	54)	**94**	55)	**9**
56)	**5**	57)	**2**	58)	**3**	59)	**3**	60)	**900**

MATHEMATICS

61)	**4**	62)	**1**	63)	**1**	64)	**2**	65)	**1**
66)	**2**	67)	**1**	68)	**2**	69)	**1**	70)	**3**
71)	**3**	72)	**3**	73)	**4**	74)	**2**	75)	**4**
76)	**1**	77)	**1**	78)	**1**	79)	**1**	80)	**1**
81)	**2**	82)	**4**	83)	**2**	84)	**486**	85)	**2**
86)	**6**	87)	**4**	88)	**338**	89)	**6**	90)	**45**

SOLUTIONS

PHYSICS

1. The peak value of the current is $I_0 = \dfrac{V_0}{\sqrt{R^2 + \dfrac{1}{\omega^2 C_2}}} = \dfrac{V_0}{\sqrt{2}R}$

When the angular frequency is changed to $\dfrac{\omega}{\sqrt{3}}$

The new peak value is $I_0 = \dfrac{V_0}{\sqrt{R^2 + \dfrac{1}{\omega^2 C_2}}} = \dfrac{V_0}{\sqrt{2}R}$

$= \dfrac{V_0}{2R} \therefore I_0 = \dfrac{I_0}{\sqrt{2}}$

2. The beautiful colours are seen on account of interference of light reflected from the upper and the lower surfaces of the thin films. Since, condition for constructive and destructive interference depends upon the wavelength of light therefore, coloured interference fringes are observed.

3. Apply nodal analysis

4. $a_{C.M} = \dfrac{F_{net}}{M_{net}}$

5.

6. Conservation of linear momentum in horizontal direction

$m_1 V_1 = (m_1 + m_2) V_2$

Coefficient of restitution = 1

$V_y \cos\theta = V_1 \sin\theta$

Component of velocity of ball along the inclined plane remain same

$V_1 \cos\theta = V_2 \cos\theta + V_y \sin\theta$

Solving

$$\frac{m_1}{m_2} = \cot^2\theta - 1$$

7. Resistivity of semiconductor decreases with temperature. The atoms of a semiconductor vibrate with large amplitudes at higher temperatures there by increasing conductivity not resistivity.

8. $E = \frac{R^2}{2a}\frac{dB}{dt}$

$$\tau = \lambda 2maEa = \lambda 2\pi a\frac{R^2}{2a}\frac{dB}{dt}a$$

$$\tau = \lambda\pi R^2\frac{dB}{dt}a$$

$$\int \tau dt = \lambda\pi R^2 a\int_{B_0}^{0} dB$$

Lt $L_f - L_i = -\lambda\pi R^2 B_0 a$

$$Iw = -\lambda\pi R^2 B_0 a$$

$$w = \frac{\lambda\pi R^2 B_0}{ma}$$

9. B due to long straight wire $= \frac{\mu_0 I_1}{2\pi r}$

Force on the current wire $= \int_{r_1-a}^{r_2-5a}\frac{\mu_0 I_1 I_2}{2\pi r}dx$

$$= \frac{\mu_0 I_1 I_2}{2\pi}\ln 5$$

10. $dV = \frac{dq}{4\pi\varepsilon_0 r} = \frac{\sigma r\frac{\pi}{2}.dr}{4\pi\varepsilon_0 r}$ $\qquad\qquad$ $\therefore V = \frac{\sigma}{8\varepsilon_0}\int_{R/2}^{R} dr$

11. Velocity of efflux $v = \sqrt{2gh}$

Velocity of top layer $= \frac{AV}{Av} = \frac{A}{A_0}\sqrt{2gh}$

$\therefore$ acceleration top layer $= \frac{d}{dx}\left[\frac{A}{Av}\sqrt{2gh}\right]$

$\frac{A}{Av}\sqrt{2g} \times \frac{1}{2\sqrt{h}} \times \frac{dh}{dt}$

$= \frac{A}{A_0}\sqrt{2g} \times \frac{1}{2\sqrt{h}} \times \frac{A}{A_0}\sqrt{2gh}$

$\left(\frac{A}{A_0}\right)^2 g$

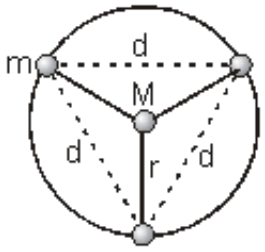
Correct option is (3)

12.

13. The distance between the orbiting stars is $d = 2r\cos 30^\circ = \sqrt{3}r$. The net inward force on orbiting stars is

$\frac{Gm^2}{d^2}\cos 30^\circ + \frac{GMm}{r^2} + \frac{Gm^2}{d^2}\cos 30^\circ = \frac{mv^2}{r}$

$\therefore G\left[\frac{m}{\sqrt{3}} + M\right] = \frac{4\pi^2 r^3}{T^2}$ or $T = 2p\sqrt{\frac{r^3}{G\left(M + \frac{m}{\sqrt{3}}\right)}}$



14. Heat current : $i = -kA\frac{dT}{dx}$

$idx = -kA\,dT$

$i\int_0^{\ell} dx = -A\alpha\int_{T_1}^{T_2} T dT$

$$\Rightarrow i\ell = -A\alpha\frac{\left(T_2^2 - T_1^2\right)}{2a}$$

$$\Rightarrow i = \frac{A\alpha\left(T_1^2 - T_2^2\right)}{2\ell}$$

15. The compressive force in the rod will be same, therefore, compressive stress will be maximum at this end. Hence, option (b) along is correct

16. $\frac{1}{2}mv^2 = mg^R$

$$\frac{mv^2}{R} = 2mg$$

$$N_{\max} = 3mg$$

17. Since the EM wave is incident normally on the reflecting wall, it will reflect back in the same direction with $\hat{i} = -\hat{i}$ and $z = -z$ creating the additional phase of π^c. We have given the equation of the incident wave, $E = E_\circ\hat{i}\cos\left(kz - \omega t\right)$.

Therefore, we can write the reflected wave equation as,

$$E = E_\circ\left(-\hat{i}\right)\cos\left(k\left(-z\right) - \omega t + \pi\right)$$

$$\Rightarrow E = -E_\circ\hat{i}\cos\left(-\left(kz + \omega t\right) + \pi\right)$$

$$\Rightarrow E = -E_\circ\hat{i}\cos\left(\pi - \left(kz + \omega t\right)\right)$$

We know the identity, $\cos\left(\pi - \theta\right) = -\cos\theta$

Using this identity in the above equation, we get,

$$\therefore E = E_\circ\hat{i}\cos\left(kz + \omega t\right)$$

18. Let l_0 be the natural length of wire and A be the cross-sectional area.

Using Hooke's law,

$\sigma\left(\text{stree}\right)\infty\in\left(\text{strain}\right)$

$$\Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{\in_1}{\in_2}$$

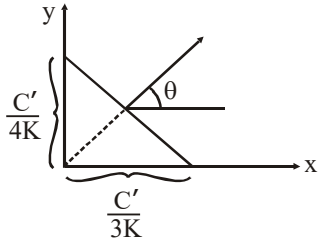
$$\Rightarrow \frac{T_1 / A}{T_2 / A} = \frac{(l_1 - l_0)/l_0}{(l_2 - l_0)/l_0}$$

$$\Rightarrow \frac{T_1}{T_2} = \frac{l_1 - l_0}{l_2 - l_0}$$

19. Equation of resultant wave is

$$Z = z_1 + z_2 = 2A\cos\left(\frac{3Kx - 4Ky}{2}\right)\cos\left(\frac{3Kx + 4Ky}{2} - \omega t\right)$$

Equation of wavefront is $\frac{3Kx + 4Ky}{2} - \omega t = C$ (C is constant)



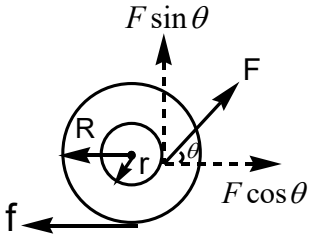
So at any given instant of time the equation of wave front will be
3Kx + 4 Ky = C′ , Wave travels perpendicular to the wave front also with increase in time both x and y should increase, as $\tan\theta = \frac{C'/3K}{C'/4K} = \frac{4}{3}$, $\theta = \tan^{-1}\frac{4}{3}$.

21. The object and two images will exist on a circle having centre at J and radius a. Also the angle made by arc joining two images forms an angle 2θ at J.
$\therefore$ Distance = $2a\sin\theta$

22. If spool is not to translate $F\cos\theta = f(1)$,If spool is not to rotate Fr = fR.. (2)

From eq. (1) and (2) we get static friction $\frac{fR}{r}\cos\theta = f$

or $\cos\theta = \frac{r}{R}$ or $\theta = \cos^{-1}\left(\frac{r}{R}\right)$



23. $H = \frac{v^2\sin^2\theta}{2g}$, $\sin 60 = \frac{H}{l}$ $\quad H = \frac{l\sqrt{3}}{2}$ $\quad \frac{l\sqrt{3}}{2} = \frac{v^2\sin^2\theta}{2g}$ $\quad U^2\sin^2\theta = 10\sqrt{3}l$

Range $= l = \dfrac{v^2 sin2\theta}{g}$ $\quad 10l = v^2 \sin 2\theta$ $\quad \dfrac{10l}{\sin 2\theta}\sin^2\theta = 10\sqrt{3}l$ $\quad \theta = \tan^{-1}\left(2\sqrt{3}\right)$

24. $Be^9(\alpha, n)c^{12}$

Taking components along X and Y-axis of momentum before and after collision

$P_\alpha = Pc\cos\theta$ $\quad 0 = Pn - Pc\sin\theta$ (or) $\quad P_\alpha^2 + P_n^2 = P_c^2 \Rightarrow 4T_\alpha + T_n = 12T_c$

$T_c = \dfrac{T_\alpha}{3} + \dfrac{T_n}{12}$ $\qquad T = K.E. = \dfrac{P^2}{2m}$

$and\ Q = T_c + T_n - T_\alpha$

25. $\dfrac{L}{7}\left[1 + \dfrac{6}{5} + \dfrac{7}{5} + \dfrac{4}{5}\right] = \dfrac{22L}{35}$

26. Due to the motion of the conductor in magnetic field, an e.m.f. is induced in it. As a result, a current flows through the conductor. According to Lenz's law, a force $Bi\ell$ (due to induced current) opposes the motion of the conductor. Let at some instant t, velocity of the conductor be v. The net accelerating force on conductor is $F = mg - Bi\ell$ …(1)

Here, induced e.m.f. $= B\ell v$ Charge on the capacitor, $q = Ce = C(B\ell v)$

Since, v is increasing, the charge and hence the current through the capacitor is also increasing. The current through capacitor is given by $i_c = \dfrac{dq}{dt} = CB\ell f\dfrac{dv}{dt}$ …(2)

From equations (1) and (2), we get $m\dfrac{dv}{dt} = mg - B\left(CB\ell\dfrac{dv}{dt}\right)\ell$

$\Rightarrow m\dfrac{dv}{dt} = mg - B^2\ell^2 C\dfrac{dv}{dt}$ $\qquad \Rightarrow \dfrac{dv}{dt}\left[m + B^2\ell^2 C\right] = mg$

$\Rightarrow a = \dfrac{dv}{dt} = \dfrac{mg}{\left(m + B^2\ell^2 C\right)}$ $\qquad \Rightarrow x(t) = \dfrac{1}{2}at^2 = \dfrac{mgt^2}{2\left(m + B^2\ell^2 C\right)} = 4$

27. $\dfrac{dg}{d\phi} = 2g_0\beta\sin\phi\cos\phi$ $\quad$ So, $d\phi = \dfrac{dg}{2g_0\beta\sin\phi\cos\phi} = 1.09\times10^{-6}\,radian = \dfrac{dg}{\beta g_0 \times \sin 2\phi}$

28. given by $\Delta E = 13.6\left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}.\right)eV$ ...(1)

This transition energy is shared between recoiling helium ion and photon

From conservation of energy. $\Delta E = jf + K.E.$ ....(2)

Where K.E. is kinetic energy of recoiling atom and hf is energy of photon

From conservation of momentum, $\vec{P}_{Photon} = -\vec{P}_{He}$ ...(3)

$$= \frac{2\Delta E}{\sqrt{1+\frac{2\Delta E}{mc^2}}+1} \qquad ...(4)$$

$$\Delta E = 13.6\left(\frac{1}{1^2}-\frac{1}{2^2}\right)eV = 40.8eV \qquad ...(5)$$

$$\frac{1}{2}mv^2 = (40.8-13.6)eV = 37.2eV \therefore \quad v = 3.1\times10^6\ m/s \quad \therefore \quad x = 6$$

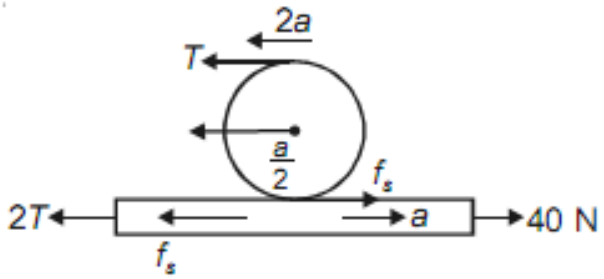
29. FBD of cylinder and block are as shown by Newton's laws

$40-2T-f_s = 8a$ (1)

$T-f_s = 4a/2$ (2)

Subtracting equation (2) from equation (1)

$40-3T = 6a$



$T = (40-6a)/3$

Also, by $\tau = I\alpha$, we get

$$T\times R + f_s \times R = I\frac{3a}{2R}$$

$$\Rightarrow T + f_s = 3Ia/2R^2$$

$$\Rightarrow (40-6a)/3 + (40-12a)/3 = 3Ia/2R^2 \Rightarrow a = 80/\left(18+9I/2R^2\right)$$

As $\tau = I\alpha = I\times(3a/2R)$

$\therefore \tau = I\alpha = I\times(3a/2R)$

$$\therefore \tau = \frac{3I}{2R}\frac{80}{\left(18+\frac{9I}{2R^2}\right)} \Rightarrow \tau = \frac{3\times80}{18+9} = \frac{80}{9}N-m$$

30. $I_{Res} = I_o\cos^2\delta/2$ $I_{Res} = \frac{3I_o}{4} \Rightarrow \delta = \pi/3 \Rightarrow \Delta x\frac{\lambda}{6} \Rightarrow x = \frac{\Delta xD}{d} = \beta/6$

CHEMISTRY

31. Maltose & Lactose are reducing sugars

32. For H: $3s = 3p = 3d \Rightarrow$ degeneracy $= 9$

33. Statement - 1 and 3 are correct

34. 1 and 3 are correct

35. $\Delta H > 0, \Delta S > 0, \quad K_2 > K_1$

36. Area $= p^{\circ} v_{\circ}$, anticlockwise $\Rightarrow W > 0$

37. $[Ni(CO)_4]$ is SP^3 and tetrahedral

38. both have non zero dipole moment

39. 1: Hex-4-en-2-one

    2: Pent-3-enoic acid

    3: 2-Butene

40. propanamide on Hofmann Bromamide degradation gives ethylamine

41. P-Toluene Sulphonyl Chloride

42. Gives Aniline

43. TACGAAGT

44. SnO_2, PbO_2 are amphoteric

45. Ga remains as Liquid upto $2000^{\circ} C$

46. Covalency = 4, o.s = +3

47. Fluorspar - CaF_2

48. BE order $Cl_2 > Br_2 > F_2 > I_2$

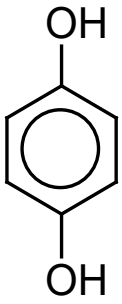
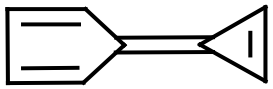
49. Fp: soln < Solvent

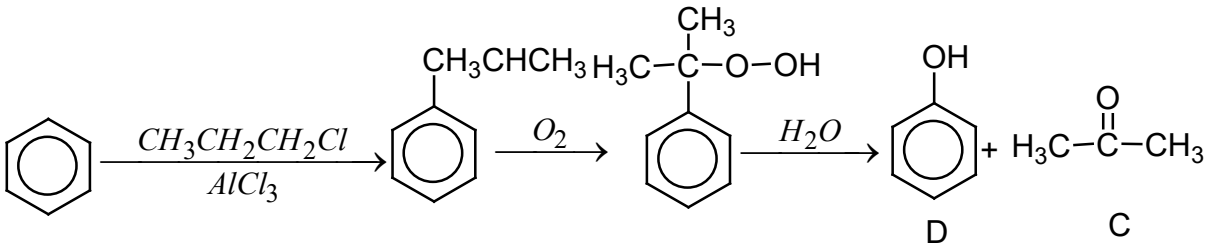
50. NCERT P/112

51. $\Delta ci = -2.3 \times 8.3 \times 298 \log^{2.5 \times 10^{-29}}$

    $= +162700 J$ =162.7 KJ =163

52. HOMO For O_2 is $\pi 2^* p_x \ \pi 2^* py \Rightarrow 2e^{-1}$

53.  and  are polar

54. 

55. $t\ 99.9 = 10t\frac{1}{2}$

56. $pH = \frac{Pka_1 + pka_2}{2} = \frac{3 - \log^{2.5} + 8}{2} = \frac{11 - 0.4}{2} = \frac{10.6}{2} = 5.3$

Nearest integer = 5.3

57. Largest $\rightarrow Mg \rightarrow 12e^-$

Smallest $\rightarrow Al^{3+} \rightarrow 10e^-$

Difference = 12 - 10 = 2

58. $MNaCl = 3 \times 58.5 = 175.5$

$m_{Sol^n} = 1250g \qquad m_{water} = 1250 - 75.5 = 1074.5$

Molality $= \frac{3}{1.0745} = 2.79$ mole kg^{-1}

59. Group2: $\left(Cu^{2+}, Pb^{2+}, Bi^{3+}, Cd^{2+}, Hg^{2+}\right)$

60. $\Delta H - T\Delta S = 0 \quad \Delta S = \frac{\Delta H}{T} = \frac{-270 \times 1000}{300}$ = - 900J

MATHEMATICS

62. $\frac{xy}{x+y} = n \Rightarrow xy - nx - ny = 0$

$\Rightarrow (x-n)(y-n)=n^2$

No. of positive solutions = No. of division of $n^2=5\times9\times13=585$

63. Solution Equation of the parabola is

$\left(y+\frac{35a}{16}\right)=\frac{a^3}{3}\left(x+\frac{3}{4a}\right)^2$ whose vertex is

$x=-\frac{3}{4a}, y=-\frac{35a}{16}\Rightarrow xy=\frac{105}{64}$

64. $4\cos^2\theta-1=4\left(1-\sin^2\theta\right)-1=3-4\sin^2\theta=\frac{\sin3\theta}{\sin\theta}$ ……..(i)

So, given expression can be written as

$=\frac{\sin27^0}{\sin9^0}\times\frac{\sin81^0}{\sin27^0}\times\frac{\sin243^0}{\sin81^0}\times\frac{\sin729^0}{\sin243^0}\left[\because By(i)\right]$

$=\frac{\sin729^0}{\sin9^0}=1$

65. $A^{43}B^{64}-B^{64}A^{43}$ is skew Symmetric

$A^{64}B^{43}-C^{43}A^{64}$ is Symmetric

66. $f(x)=\frac{4^x}{\left(4^x+4\right)}.....(i)$

Replacing x by 2-x

$f(2-x)=\frac{4^{2-x}}{\left(4^{2-x}+4\right)}=\frac{4}{\left(4^x+4\right)}.....(ii)$

Adding the equation (i) & (ii)

$\Rightarrow f(x)+f(2-x)=1.....(iii)$

67. $\underset{x\to2}{Lt}\left[\frac{(x-2)\left(x^2+2x+4\right)}{x(x-2)(x+2)}-\left(\frac{\sqrt{x}}{\sqrt{x}-\sqrt{2}}-\frac{\sqrt{2}}{\sqrt{x}-\sqrt{2}}\right)^{-1}\right]$

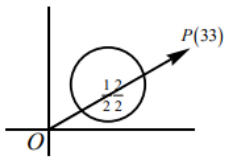
$$\lim_{x\to 2}\left[\frac{x^2+2x+4}{x(x+2)}-1\right]=\frac{12}{8}-1=\frac{1}{2}$$

68. $\dfrac{z-i}{z-1}=\dfrac{x+iy-i}{x+iy-1}$

$$=-\frac{x(x-1)+y(y-1)+i(-y-x+1)}{(x-1)^2+y^2}$$

Put = 0 $\quad\Rightarrow x^2-x+y^2-y=0$

$$\Rightarrow\left(x-\frac{1}{2}\right)^2+\left(y-\frac{1}{2}\right)^2=\frac{1}{2}\qquad C=\left(\frac{1}{2},\frac{1}{2}\right)$$



69. $Q=PAP^T$

$P^TQP=A\qquad\left(\therefore PP^T=I\right)$

$P^TQ^{2005}P=AP^TQ^{2004}P$

$=A^2P^TQ^{2003}P=A^3P^TQ^{2002}P=.....$

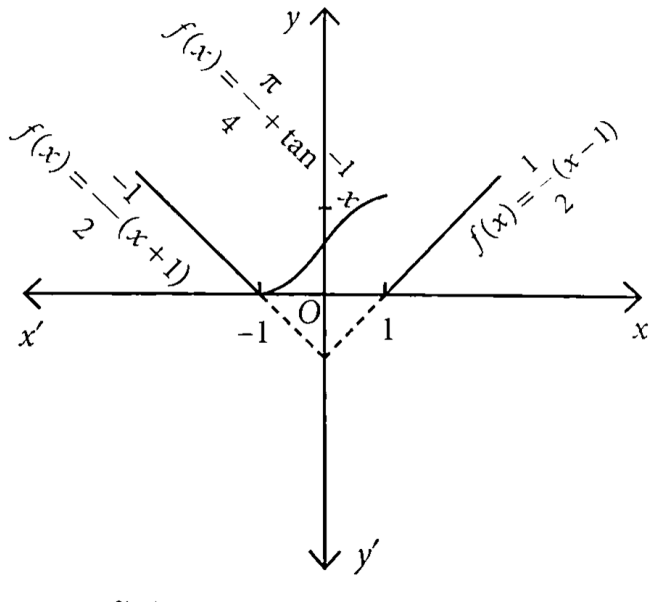
$=A^{2004}P^T(QP)=A^{2004}P^T(PA)$

$\left(\because Q=PAP^T\Rightarrow QP=PA\right)\qquad=A^{2005}=\begin{bmatrix}1 & 2005\\ 0 & 1\end{bmatrix}$

70. equation of circle is $x^2+y^2-\frac{4}{5}x-\frac{44}{5}y+\frac{84}{5}=0$

71. Here, $f(x)=\begin{cases}\frac{\pi}{4}+\tan^{-1}x & |x|\le 1\\ \frac{1}{2}(|x|-1), & |x|>1\end{cases}$

Thus from the graph it is clear that $f(x)$ is continuous on R – {1} and differentiable on R – {-1,1}.



72. We have,

$f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5$

$f'(x) = \dfrac{4}{x-1} - 4x + 4$ .....(i)

And $f''(x) = \dfrac{-4}{(x-1)^2} - 4$ .....(ii)

Now, $f'(x) > 0$ for increasing and $f'(x) < 0$ for decreasing

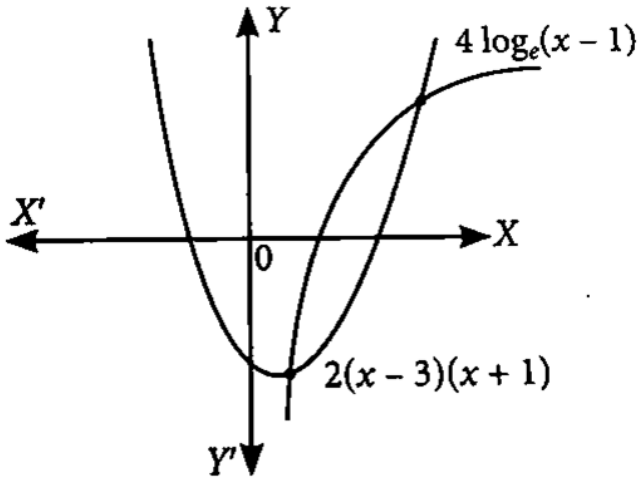
$\Rightarrow \dfrac{2x - x^2}{x-1} > 0, \dfrac{2x-x^2}{x-1} < 0$ [Using (i)]

$\Rightarrow x \in (-\infty, 0) \cup (1,2), x \in (0,1) \cup (2,\infty)$

So, option (1) is correct.

If $f(x) = -1 \Rightarrow 4\log_e(x-1) = 2x^2 - 4x - 6$

$\Rightarrow 4\log_e(x-1) = 2(x-3)(x+1)$



From the graph, it is clear that the above equation has exactly 2 solutions.

(3) is false.

(4) is correct.

73. By substituting $x=\frac{1}{t}$ and $dx=-\frac{1}{t^2}dt$ in equation (1),

$$I=-\int_{2}^{1/2}\frac{\tan^{-1}\frac{1}{t}}{\frac{1}{t}}\cdot\frac{1}{t^2}\cdot dt$$

$$I=-\int_{2}^{1/2}\frac{\tan^{-1}\frac{1}{t}}{t}dt=\int_{1/2}^{2}\frac{\cot^{-1}t}{t}dt=\int_{1/2}^{2}\frac{\cot^{-1}x}{x}dx\ .....(2)$$

On adding equation (1) and (2), we have

$$2I=\int_{1/2}^{2}\frac{\tan^{-1}x+\cot^{-1}x}{x}dx;2I=\frac{\pi}{2}\int_{1/2}^{2}\frac{dx}{x}=\frac{\pi}{2}\left[\log x\right]_{1/2}^{2}$$

$$2I=\frac{\pi}{2}\left[\log_e 2-\log_e\frac{1}{2}\right]=\pi\log_e 2$$

$$\Rightarrow 2I=\pi\log_e 2\ \therefore\ I=\frac{\pi}{2}\log_e 2$$

Hence option (4) is correct answer.

74. Shortest distance between lines

$\frac{x-\lambda}{0}=\frac{y-3}{4}=\frac{z+6}{1}$ and $\frac{x+\lambda}{3}=\frac{y}{-4}=\frac{z-6}{0}$ is 13

So, $\left|\frac{\begin{vmatrix}\lambda+\lambda & 3-0 & -6-6\\ 0 & 4 & 1\\ 3 & -4 & 0\end{vmatrix}}{\sqrt{(0+4)^2+(3-0)^2+(0-12)^2}}\right|=13$

$\Rightarrow\left|\frac{\begin{vmatrix}2\lambda & 3 & -12\\ 0 & 4 & 1\\ 3 & -4 & 0\end{vmatrix}}{13}\right|=13$

$\Rightarrow\left|2\lambda(0+4)-3(-3)-12(0-12)\right|=13\times13$

$\Rightarrow\left|8\lambda+9+144\right|=169\Rightarrow\left|8\lambda+153\right|=169$

$\Rightarrow 8\lambda+153=169$ and $8\lambda+153=-169\Rightarrow\lambda=2,\frac{-161}{4}$

$\therefore 4\left|\sum_{\lambda\in S}\lambda\right|=4\left|\frac{-161}{4}+2\right|=4\left|\frac{-153}{4}\right|=153$

76. $\alpha^2-6\alpha-2=0$, multiply by α_8 on both sides

$\Rightarrow a^{10}-8\alpha^9-2\alpha^8=0$

$\Rightarrow\beta^{10}-8\beta^9-2\beta^8=0$

$\therefore\left(\alpha^{10}-\beta^{10}\right)-8\left(\alpha^9-\beta^9\right)-2\left(\alpha^8-\beta^8\right)=0$

$\Rightarrow a_{10}-8a_9-2a_8=0$

$\Rightarrow\frac{a_{10}-2a_8}{4a_9}=2$

77. $x^3+x-1=9\Rightarrow x=2$

$\Rightarrow g(9)=2$

$g(f(x))=x \Rightarrow g'(f(x).f'(x))=1$

$\Rightarrow g'(f(x))=\dfrac{1}{f'(x)}=\dfrac{1}{f'(2)}=\dfrac{1}{13}$

Put $f(x)=9$

$\Rightarrow x=2$

$\Rightarrow \dfrac{g(9)}{g'(9)}=26$

78. $\dfrac{\frac{4}{5}\times\frac{1}{4}\times\frac{1}{4}+\frac{4}{5}\times\frac{3}{4}\times\frac{3}{4}}{\frac{4}{5}\times\frac{1}{4}\times\frac{1}{4}+\frac{4}{5}\times\frac{3}{4}\times\frac{3}{4}+\frac{1}{5}\times\frac{3}{4}\times\frac{1}{4}+\frac{1}{5}\times\frac{1}{4}\times\frac{3}{4}}=\dfrac{20}{23}$

79. $A=\{x\in\mathbb{R}:[x+3]+[x+4]\le 1\}$,

$2[x]+7\le 1$

$2[x]\le -6$

$[x]\le -3 \Rightarrow x<-2 \qquad ...(A)$

$B=\left\{x\in R:3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$,

$3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}$

$3^{2x-3}\left(\frac{\frac{1}{10}}{1-\frac{1}{10}}\right)^{x-3}<3^{-3x}$

$\Rightarrow\left(\frac{1}{9}\right)^{x-3}<3^{-5x+3}$

$\Rightarrow 3^{6-2x}<3^{3-5x}$

$\Rightarrow 6-2x<3-5x \Rightarrow 3x<-3 \Rightarrow x<-1$

80. $S_n = 1!+2!+3!+4!+5!+6!+7I$ where I be an integer.

$S_n = 873+7I$

$\frac{S_n}{7} = 124.7 + I \Rightarrow \left[\frac{S_n}{7}\right] = 124 + I$

$\therefore 7\left[\frac{S_n}{7}\right] = 868+7I \Rightarrow S_n - 7\left[\frac{S_n}{7}\right] = (873+7I)-(868+7I) = 5$

Now, $\sin^{-1}(\sin 5) = \sin^{-1}(\sin(5-2\pi)) = 5-2\pi \Rightarrow (P)$

$\cos^{-}(\cos 5) = \cos^{-1}(\cos(2\pi-5)) = 2\pi-5 \Rightarrow (Q)$

$\tan^{-1}(\tan 5) = \tan^{-1}(\tan(5-2\pi)) = 5-2\pi \Rightarrow (P)$

and $\cot^{-1}(\cot 5) = \cot^{-1}(\cot(5-\pi)) = 5-\pi \Rightarrow (S)$

81. $x - \frac{x^2}{2} + \frac{x^3}{4} + ... = x^2 - \frac{x^4}{2} + \frac{x^6}{4} + ...$

$\Rightarrow \quad \frac{x}{1-\left(-\frac{x}{2}\right)} = \frac{x^2}{1-\left(-\frac{x^2}{2}\right)}$

$\Rightarrow \quad \frac{2x}{2+x} = \frac{2x^2}{2+x^2} \Rightarrow \quad x(x-1)=0$

$\Rightarrow \quad x = 0,1$ but $0<|x|<\sqrt{2}$

$\Rightarrow \quad x = 1$

82. $\det A = (2k-1)\left(4k^2-1\right) + 2\sqrt{k}\left(4k\sqrt{k}+2\sqrt{k}\right) + 2\sqrt{k}\left(4k\sqrt{k}+2\sqrt{k}\right)$

$= (2k-1)\left(4k^2-1\right) + 8(2k+1)k$

$= (2k+1)\left((2k-1)^2+8k\right) = (2k+1)^3$

$\therefore \det B = 0$ since B is skew symmetric matrix of order 3.

$\therefore \det(adjB) = (\det B)^2 = 0$

$\therefore (2k+1)^{12} = 10^{12}$

$\Rightarrow 2k+1 = 10 \Rightarrow k = 4.5$

83. $\dfrac{dy}{dx} + \dfrac{2x}{x-1}y = \dfrac{1}{(x-1)^2}$

$I.F. = e^{\int \frac{2x}{x-1}dx} = e^{2\int\left(1+\frac{1}{x-1}\right)dx} = e^{2x+2\log(x-1)} = e^{2x}(x-1)^2$

Solution is $e^{2x}(x-1)^2 y = \dfrac{e^{2x}}{2} + c$

$e^0(0-1)^2 1 = \dfrac{e^0}{2} + c \Rightarrow c = 1 - \dfrac{1}{2} = \dfrac{1}{2} \therefore e^{2x}(x-1)^2 y = \dfrac{e^{2x}}{2} + \dfrac{1}{2}$

$e^4(2-1)^2 y(2) = \dfrac{e^4+1}{2}$

$y(2) = \dfrac{e^4+1}{2e^4}$

84. $(1+x)^9\left(1+x^2\right)^9 = \left(1+9c_1x+9c_2x^2+....\right)\left(1+9c_1x^2+9c_2x^4+...\right)$

$\therefore$ Coefficient of $x^4 = 9c_2 + 9c_2.9c_1 + 9c_4 = 486$

85. only $(-2\pi, 0)$, $(0,0)$ satisfy the given equations

86. $2^m - 2^n = 96$

87. We have, $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$

$\Rightarrow \sum_{i=1}^{18} X_i = 18(\alpha+2)$ and $\Rightarrow \sum_{i=1}^{18} X_i^2 - 2\beta\sum_{i=1}^{18} X_i + 18\beta^2 = 90$

$\Rightarrow \sum_{i=1}^{18} X_i^2 = 90 - 18\beta^2 + 36\beta(\alpha+2)$

Since $\sigma^2 = 1$ (given)

$$\Rightarrow \frac{\sum_{i=1}^{18} X_i^2}{18} - \left(\frac{\sum_{i=1}^{18} X_i}{18} \right)^2 = 1$$

$$\Rightarrow \frac{1}{18}\left(90 - 18\beta^2 + 36\beta(\alpha+2)\right) - \left(\frac{18(\alpha+2)}{18}\right)^2 = 1$$

$$\Rightarrow 90 - 18\beta^2 + 36\beta(\alpha+2) - 18(\alpha+2)^2 = 18$$

$$\Rightarrow 5 - \beta^2 + 2\alpha\beta + 4\beta - \alpha^2 - 4\alpha - 4 = 1$$

$$\Rightarrow \alpha^2 + \beta^2 - 2\alpha\beta - 4\beta + 4\alpha = 0$$

$$\Rightarrow (\alpha-\beta)^2 + 4(\alpha-\beta) = 0 \Rightarrow (\alpha-\beta)\left[(\alpha-\beta)+4\right] = 0$$

$$\Rightarrow (\alpha-\beta) = 0 \, or \, -4$$

$\therefore |\alpha - \beta| = 4$ [$\therefore \alpha, \beta$ are distinct real numbers]

88. Point of intersection of $l_1 : 3y - 2x = 3$ and $l_2 : x - y + 1 = 0$ is $P(0,1)$,

Which lies on $l_3 : \alpha x + \beta y + 17 = 0$

$\beta = -17$

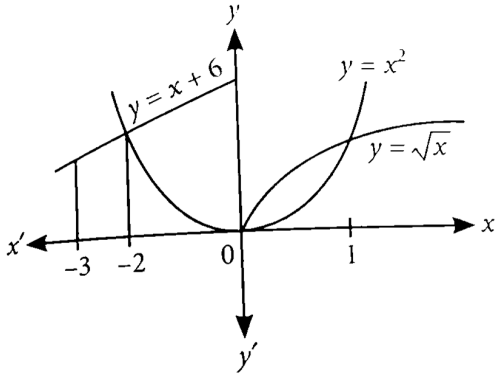
Consider a random point $Q(-1,0)$ on $l_2 : x - y + 1 = 0$.

Image of point Q about $l_1 : 2x - 3y + 3 = 0$ is $Q'\left(\frac{-17}{13}, \frac{6}{13}\right)$

which can be calculated by formula, $\frac{x-(-1)}{2} = \frac{y-0}{-3} = \frac{-2(-2+3)}{13}$

Now, Q' lies on $l_3 : \alpha x + \beta y + 17 = 0$ $\therefore \alpha = 7$

89.



Area bounded by $y = f(x)$ and x-axis,

$$A = \int_{-3}^{-2} (x+6)\,dx + \int_{-2}^{0} x^2\,dx + \int_{0}^{1} \sqrt{x}\,dx$$

$$= \left[\frac{x^2}{2} + 6x\right]_{-3}^{-2} + \left[\frac{x^3}{3}\right]_{-2}^{0} + \left[\frac{x^{3/2}}{3/2}\right]_{0}^{1} = \frac{7}{2} + \frac{8}{3} + \frac{2}{3} \qquad \therefore A = \frac{41}{6}$$

90. Let $\vec{c} = \lambda\left(\vec{b} \times \left(\vec{a} \times \vec{b}\right)\right) = \lambda\left(\left(\vec{b} \cdot \vec{b}\right)\vec{a} - \left(\vec{b} \cdot \vec{a}\right)\vec{b}\right)$

$= \lambda\left(5\left(-\hat{i} + \hat{j} + \hat{k}\right) + 2\hat{i} + \hat{k}\right) = \lambda\left(-3\hat{i} + 5\hat{j} + 6\hat{k}\right)$

Also, $\vec{a} \cdot \vec{c} = 7 \Rightarrow 3\lambda + 5\lambda + 6\lambda = 7$

$\Rightarrow 14\lambda = 7 \Rightarrow \lambda = \frac{1}{2}$